

Beyond Keywords

Evidence-based candidate ranking for the Senior AI Engineer shortlist

100k

candidate records
processed offline

CPU

no GPU and no hosted
API

226.9s

final local frozen-tag
runtime

Central release claim

Career evidence outranks
skill lists, explanations stay
grounded, and output order
remains deterministic: score
descending, then
candidate_id ascending.

The release keeps ranking logic frozen, regenerates the final CSV from the tagged code path, validates the top 100 explanations, and reproduces the same shortlist offline in Docker.

Keyword filters blur real ranking depth with copied buzzwords

The challenge is not counting AI terms. It is separating candidates who built ranking systems from candidates who only mention them.

Skill-list inflation

- Raw skills are easy to copy
- Advanced IR terms can appear without supporting work history
- Uncorroborated claims should not dominate ranking

Title ambiguity

- Senior titles can hide weak retrieval depth
- Research-heavy roles may not imply shipped ranking systems
- Descriptions matter more than labels

Recruiter noise

- Availability matters, but only within bounds
- Sparse profiles need conservative explanations
- Determinism matters for reviewer trust

The shortlist favors retrieval, ranking, evaluation, and product delivery evidence

Retrieval and ranking depth

- Search relevance
- Recommendation systems
- Matching engines
- Learning-to-rank or ranking ownership

Evaluation and shipping

- Offline metrics or A/B tests
- Production deployment
- Monitoring and iteration
- User-facing impact

Practical engineering

- Python services and APIs
- Data or search infrastructure
- Recent coding evidence
- Cross-functional product execution

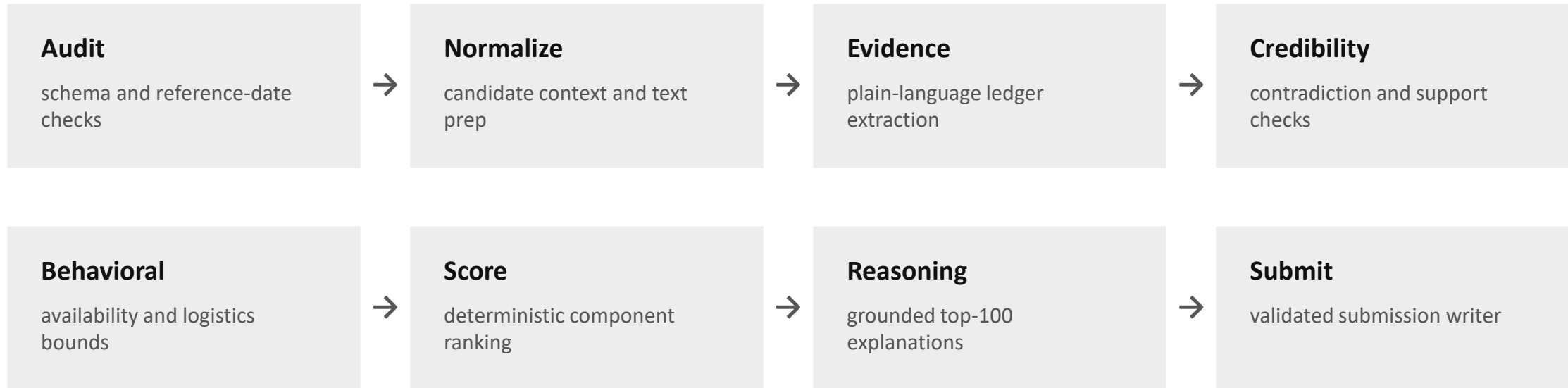
This keeps the role definition tied to shipped systems instead of broad AI adjacency.

Career history outranks summaries, and summaries outrank skill lists

1	career_history.description	Highest-signal proof of what the candidate actually built
2	career_history.title	Useful only when paired with the description
3	profile.summary / headline	Good support, weaker than work history
4	skills / assessments	Lowest-trust source unless corroborated

This ordering is the main defense against resumes that list modern tools without showing ranking-relevant ownership.

The pipeline persists every stage and only resumes when fingerprints still match



Each stage writes artifacts, hashes, and elapsed times under runs/<run_id>. Resume mode skips only when input, config, and upstream fingerprints remain unchanged.

Plain language matters more than exact buzzwords

Recognized positive evidence

- search relevance
- recommendation systems
- personalized feed
- marketplace ranking

Operational evidence

- A/B evaluation
- production deployment
- monitoring
- Python services

Scoring summary

- base_fit_score
- × credibility_multiplier
- × availability_multiplier
- × 100

A candidate can rank well by describing real ranking or retrieval work even if their wording is not a perfect keyword match.

Credibility and corroboration defenses keep copied AI language from dominating

Corroboration

- Skills alone are weak evidence
- Work history must support advanced claims
- Career evidence can outweigh keyword-heavy profiles

Contradictions

- Title and description mismatches lower credibility
- Sparse support reduces trust
- Penalties are explicit and deterministic

Bounded modifiers

- Availability range is 0.72 to 1.08
- Behavioral signals separate close candidates
- They do not rescue weak technical fit

Availability helps shortlist triage, but only inside a narrow multiplier band

0.72

1.08



shortlist triage zone

Signals used

- notice period
- last-active recency
- location logistics

What it changes

- ordering among similar candidates
- recruiter urgency
- practical interview viability

What it cannot do

- override weak ranking evidence
- invent missing experience
- replace human review

The release explanation layer changed wording without changing scores or shortlist order

100/100

top-100 style lint
passed

100/100

top-100 grounding
passed

0

first-person recruiter
leaks in final run

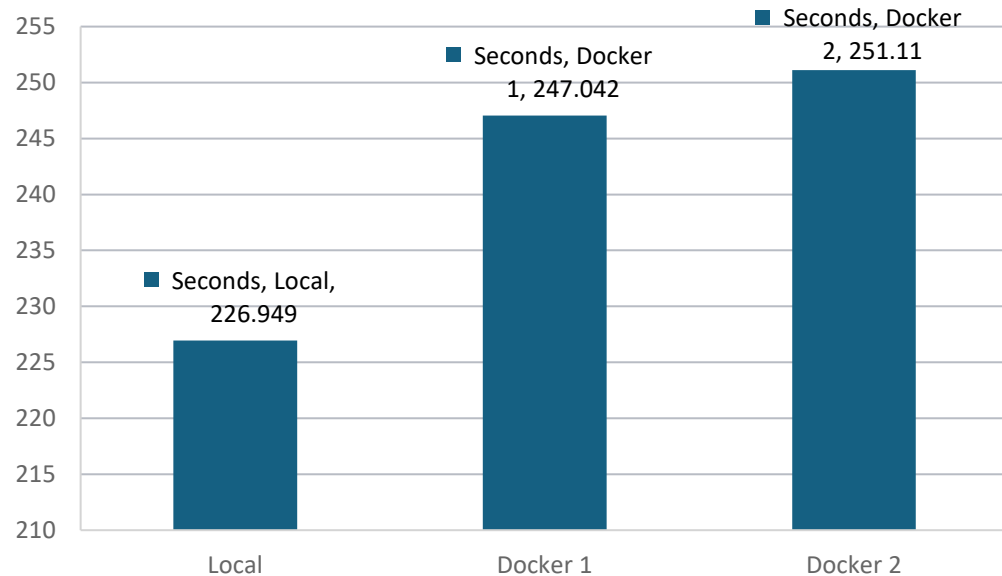
true

top-100 IDs, ranks,
scores unchanged

The Step 3B patch only neutralized first-person source phrasing such as 'our product' into recruiter-facing narration. The final local release run preserved the approved score CSV hash and the exact top-100 candidate IDs, ranks, and scores.

Reasoning is selected from source-grounded evidence IDs, linted, and then checked again for grounding before release.

The release stays offline and reproducible across local and Docker runs



3982 MB

local peak memory

7132 MB

Docker verification peak
memory

false

hosted API during ranking

true

local and Docker
submission hash match

Docker remained under the official 300-second limit but inside the project warning band, so a second network-disabled verification run was executed and matched the local output.

The recruiter sandbox uses the same ranking modules on small JSON or JSONL samples

Redrob Candidate Ranking Sandbox

Offline, CPU-only demo that runs the same ranking, evidence, credibility, behavioral, scoring, and reasoning modules as `rank.py`.

This sandbox is for small samples only. Samples under 100 candidates cannot be validated as official competition submissions.

Upload candidate JSON or JSONL

demo_c...dates.jsonl7.7KB

+

Ranked 5 candidates.

Download ranked CSV

candidate_id	rank	score	credibility_modifier	availability_modifier	reasoning
CAND_DEMO_0002	1	43.816068	0.92	0.9984	Built recommendation services, ran A/B evaluation, and improved matching quality fo
CAND_DEMO_0005	2	22.447802	0.92	1.0006	Shipped product features, Python services, and practical engineering work for user-fa
CAND_DEMO_0001	3	19.564264	0.74	1.0325	Built a semantic search feature for an internal knowledge base of 500K documents. Pr
CAND_DEMO_0004	4	17.319891	0.92	0.9164	Built backend services with some applied ML support but limited retrieval depth in the
CAND_DEMO_0003	5	6.266994	0.92	0.9698	Built experimentation frameworks that supported the product team's A/B tests. Explic

> Rank 1 - CAND_DEMO_0002

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Upload candidate JSON or JSONL

Upload

200KB per file - JSON, JSONL

Try `sample_data/demo_candidates.json` or `sample_data/demo_candidates.jsonl`.

- upload JSON or JSONL
- rank up to 100 candidates
- download ranked CSV
- marked as non-official for small samples

The app surfaces rank, score components, evidence excerpts, modifiers, and the same recruiter-facing reasoning generated by the production pipeline.

This system supports recruiter triage, not final hiring decisions

Limits

- depends on profile quality
- does not infer missing evidence
- no claims about hiring outcomes

Responsible use

- human review remains required
- top-ranked profiles should be spot-checked
- availability is secondary to role fit

Practical next step

- add recruiter feedback loop after the challenge
- keep offline guarantees
- reassess only with evidence-backed changes